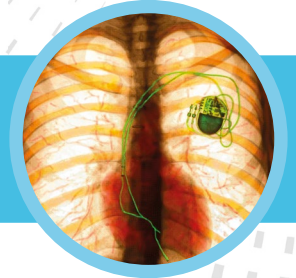


GLOSSARY



ACCELERATION

An increase or decrease in an object's speed due to a force being applied to it.

ACID

A reactive chemical that has a pH less than 7 (*see* pH). Vinegar and lemon juice are weakly acidic.

AERODYNAMICS

The science of how objects move through the air. The four aerodynamic forces are thrust, drag, lift, and weight.

ALKALI

A reactive chemical that has a pH greater than 7 (*see* pH). Soap and bleach are alkaline.

AIR PRESSURE

The force of air molecules pushing against a surface or container.

ALGAE

Simple, plantlike organisms that live in water and make food using energy from sunlight.

ANTIBODY

A protein (*see* protein) that sticks to microbes such as bacteria, stopping them from being harmful or flagging them for destruction by the body's own white blood cells. When your body encounters a new microbe, it learns to make antibodies, making you immune to it if it enters your body again.

ARTIFICIAL INTELLIGENCE

A technology that enables computers to mimic the human behavior of learning and reasoning. Most artificial intelligence are designed for analyzing data, recognizing patterns, and simulations.

ATMOSPHERE

The layers of gas that surround Earth, held by gravity.

ATOM

A tiny particle of matter. An atom is the smallest part of an element that can exist.

BACTERIA

Microscopic, single-celled organisms with no cell nuclei. Bacteria are the most abundant organisms on Earth.

BATTERY

An energy-storing device that produces an electric current when connected to a circuit.

BIOLOGY

The study of living things. It studies how plants and animals interact with their surroundings. It includes fields of study such as botany, zoology, and microbiology.

BIOME

A region with the same climate, vegetation, and animals. The same biome can be found on different continents. Some of the biomes around the world include tundra, temperate grassland, and tropical forest.

BLOOD VESSEL

Any tube that carries blood through your body. There are three main types of blood vessel: arteries, veins, and capillaries.

BRAIN

The organ inside the head that controls the whole body. This vital part of the body's nervous system allows you to sense, move, think, reason, and understand emotions.

CARBON

A nonmetallic element that is a key component of important chemicals in the body such as proteins and DNA.

CARBON DIOXIDE

A colorless, odorless gas. An increasing amount of this gas in the atmosphere is causing global warming.

CELL

The basic unit from which all living organisms are made.

CHEMICAL

An element, or a substance made of more than one element. A chemical is pure or the same all the way through—it is not a mixture. Water, iron, and oxygen are all chemicals.

CHEMICAL REACTION

A process in which atoms are rearranged to form at least one new substance.

CHEMISTRY

The study of matter and elements, and how elements react when mixed together.

CIRCUIT

A path that electricity flows around. All electrical devices have circuits inside them.

CONDUCTOR

A substance through which heat or electric current flows easily.

CORAL

A simple organism that lives on the seabed in big colonies.

CRYSTAL

A solid substance with a highly-ordered shape. Diamonds and salt grains are crystals.

CURRENT

A flow of a substance. An electric current is a flow of electrons. An ocean current is a flow of water in the ocean, driven by the wind or by differences in water density caused by temperature or salt content.

CYTOPLASM

The jellylike fluid inside a cell membrane, which makes up most of the material inside a cell.

DATA

Information, such as facts and statistics, collected for reference or analysis.

DEFORMATION

A change in an object caused by forces. When an object can't move, the forces can either change the shape of the object or cause it to break.

DENSITY

The mass of a solid, liquid, or gas per unit of volume. A dense material has lots of atoms packed closely together. Less dense objects float in more dense fluids. Wood can float in water, because it is less dense than water.

DNA

Short for "deoxyribonucleic acid," a long, thin, double-helix-shaped molecule found in the cells of all living organisms. It carries genetic code—the instructions for how a living thing will look and function.

DRAG

A force that acts on an object moving through air or water. Drag always acts in the opposite direction to the object's direction of motion.